INCH-POUND
MIL-S-24187/2B(SH)
22 March 1991

SUPERSEDING
MIL-S-24187/2A(SH)
9 September 1983

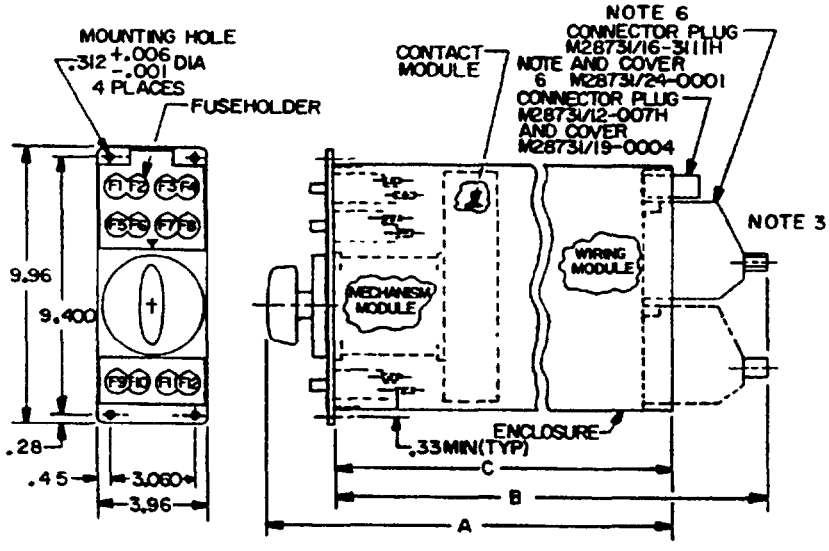
MILITARY SPECIFICATION SHEET

SWITCH ASSEMBLIES, LINEAR MOVEMENT, MANUAL AND REMOTE OPERATION, STYLE 3 BLS

This specification sheet is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-S-24187.

INCHES	MM
.001	0.03
.006	0.15
.03	0.76
.09	2.29
.12	3.05
.28	7.11
.312	7.93
.33	8.38
.45	11.43
3.060	77.72
3.960	100.58
8.44	214.38
9.40	238.76
9.960	252.98
11.12	282.45
11.94	303.28
12.82	325.63
14.00	355.60
14.62	371.35
16.32	414.53
16.68	423.67
18.38	466.85



(MANUAL SHOWN)

FIGURE 1. *Style BLS switch assembly*

AMSC N/A FSC 5930

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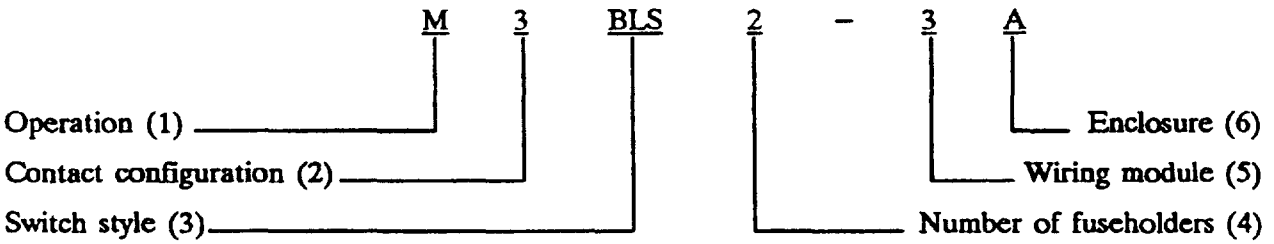
Dimensions	MBLS (with fuseholders)	MBLS (without fuseholders)	RBLS
A	14.62 ± 0.09	11.12 ± 0.09	16.68 ± 0.09
B	16.32 ± 0.12	12.82 ± 0.12	18.38 ± 0.12
C	11.94 ± 0.03	8.44 ± 0.03	14.00 ± 0.03

Notes:

1. Dimensions are in inches.
2. Unless otherwise specified, tolerances are plus or minus 0.02 (0.51 mm) for two-place decimals and plus or minus 0.005 (0.13 mm) for three-place decimals.
3. Connector cover, piece M28731/24-0001, may be rotated 180 degrees to accept cable from bottom.
4. Supplied only when specified.
5. Metric equivalents (to the nearest 0.01 mm) are given for general information only and are based upon 1 inch + 25.4 mm.

FIGURE 1. *Style BLS switch assembly - Continued.*

Type designation:

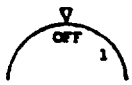
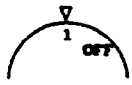
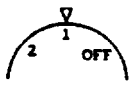
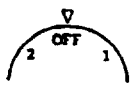
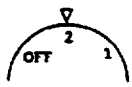
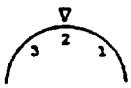
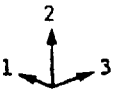
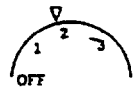
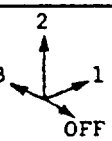
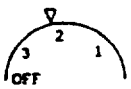
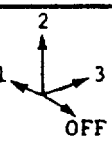


1. Switch assemblies shall be remote operated with manual override, designated R, or manually operated, designated M.
2. Contact configuration shall be type 3 (see figure 2).
3. Switch style shall be designated by the letters BLS.
4. Number of double circuit fuseholders, installed, when provided for, shall be indicated by a single digit, 0 to 6 (0 for RBLS).
5. Wiring module circuits shall be indicated by a single digit (1, 2 or 3) to accomplish the switching actions specified in table I.
6. Length of enclosure shall be indicated by a single letter.

   A – Manual switch with fuseholders 12.0 inches (304.8 mm)
   B – Manual switch with no fuseholders 8.50 inches (215.9 mm)
   C – Remote switch, no extra fuseholders 14.0 inches (355.6 mm)

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TABLE I. *Switch selection.*

Wiring module assy	Switch config-uration	Stop position[1]	Dial engraving[2]	Handle position[3]	Wired contacts[4]	To SW conn no[5]	Non-wired contacts[6]
201	1-OFF	1 & 3			( )A LOAD 1 ( )X INPUT	JC JB	( )D ( )C ( )B OFF
202	OFF-1	1 & 3			( )B LOAD 1 ( )X INPUT	JD JB	( )D ( )C ( )A OFF
203	OFF-1-2	1 & 2			( )C LOAD 2 ( )B LOAD 1 ( )X INPUT	JE JD JB	( )D ( )A OFF
204	1-OFF-2	1 & 2			( )C LOAD 2 ( )A LOAD 1 ( )X INPUT	JE JC JB	( )D ( )B OFF
205	1-2-OFF	1 & 2			( )B LOAD 2 ( )A LOAD 1 ( )X INPUT	JD JC JB	( )D ( )C OFF
206	1-2-3	1 & 2			( )C LOAD 3 ( )B LOAD 2 ( )A LOAD 1 ( )X INPUT	JE JD JC JB	( )D
206	3-2-1-OFF	1			( )C LOAD 1 ( )B LOAD 2 ( )A LOAD 3 ( )X INPUT	JE JD JC JB	( )D OFF
206	1-2-3-OFF	1			( )C LOAD 1 ( )B LOAD 2 ( )A LOAD 3 ( )X INPUT	JE JD JC JB	( )D OFF

[1]Stops shall be adjustable to limit switch handle rotation to the positions indicated or to any consecutive positions within the maximum rotation indicated.

[2]Reference position of a BLS switch is considered to be that in which the flat on the handle shaft is up. The mark ∇ and the dial positions shown are used to designate the reference position.

[3]Normal position of a BLS switch is the position which coincides with normal circuit conditions in any application. The handle shall be adjustable to point upward in the normal position or to any specified position, in 45 degree increments from upward. The handle and rotary description plate rotate.

[4]See figures 2 and 3 for complete description of contact identification. Parentheses ( ) shall be replaced by appropriate pole group letter.

[5]For switch assemblies using wiring modules (see figure 5 for connector identification)

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REQUIREMENTS:

Construction: Style BLS switch parts, modules and assemblies shall be constructed in accordance with Drawing 803-1853221.

Characteristics: Characteristics shown in table II shall apply to the style BLS switch assembly.

TABLE II. *Characteristics.*

Characteristics	Value
Total number of poles available	98
Wired load switching poles	93
Number of positions:	2 to 4
Break-before-make	Yes
Make-before-break	No
Electrical rating and endurance loading:	
4 Amperes, 115 V, ac resistive	20,000 Operations
3 Amperes, 115 V, ac, (0.3 p.f.), 60 Hz	20,000 Operations
3 Amperes, 115 V, ac, (0 1 p.f.), 400 Hz	20,000 Operations
4 Amperes, 28 V, dc, resistive	20,000 Operations
25 Microamperes, 1.5 V, dc	20,000 Operations
Remote operation	
Transfer time·	
Minimum	0 030 seconds
Maximum	0.125 seconds
Actuating time:	
1 Position change	0.25 seconds
3 Position change (one direction)	1 10 seconds
Current drain.	
At 115 V, 3 ϕ, 60 Hz	1 Ampere rms per phase max (0 70 Amp, typ)
Contact resistance	
At 4 amperes	0.025 ohm
At 2.5 microamperes	0.05 ohm
Total circuit resistance:	
At 4 amperes	0 080 ohm
At 25 microamperes	0.115 ohm
Torque – shaft rotation maximum – Initial	40 pound-inches
– After shock	40 pound-inches
– stop strength	150 pound-inches
Maximum weight:	
Enclosed with wiring module	20 pounds
Without enclosure and wiring module	14 pounds
Temperature and humidity (for rotational torque and endurance tests)	
Minimum	Plus 40 °F (plus 4 °C)
Maximum	Plus 149 °F (plus 65 °C) and 40 to 50 percent rh
Shock	MIL-S-901, grade A, class I, type C
Vibration	MIL-STD-167-1, type I, except omit variable frequency
Temperature rise	30 °C

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Switch assembly modules: Style BLS manually operated switch assemblies shall consist of the following modules as detailed on Drawing 803-1853221.

Switch assembly
- Front module (MBLS only)
- Mechanism module
- Switch contact module

Wiring module
Enclosure

Remote operated, style BLS switch assemblies shall have, in addition to the above, the

Remote control module/mechanism module (associated circuitry)
Control circuit module

Switch function: Style BLS switch assemblies shall provide 93 4-position poles each consisting of five stationary contacts X, A, B, C and D and a movable contact arranged as shown on figure 2. A stop shall prevent transfer of the movable contact into position X. Additional stops (removable) shall be provided which, when inserted as specified in table III, will limit movement to two or three consecutive positions. Stops shall be as detailed on Drawing 803-1853221 and shall be accessible from the front of the switch assembly, after removal of the handle and rotary description plate.

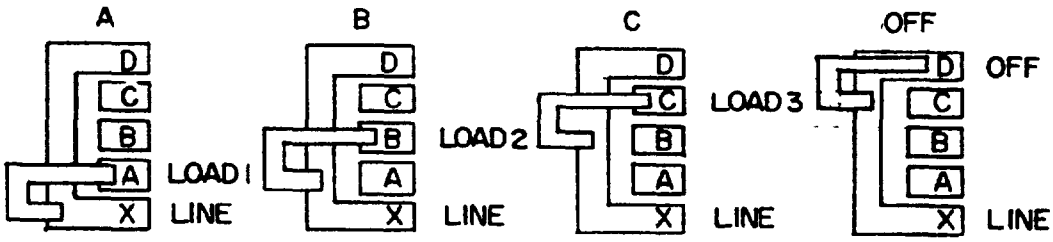


FIGURE 2. *Schematic typical operation of 1 pole.*

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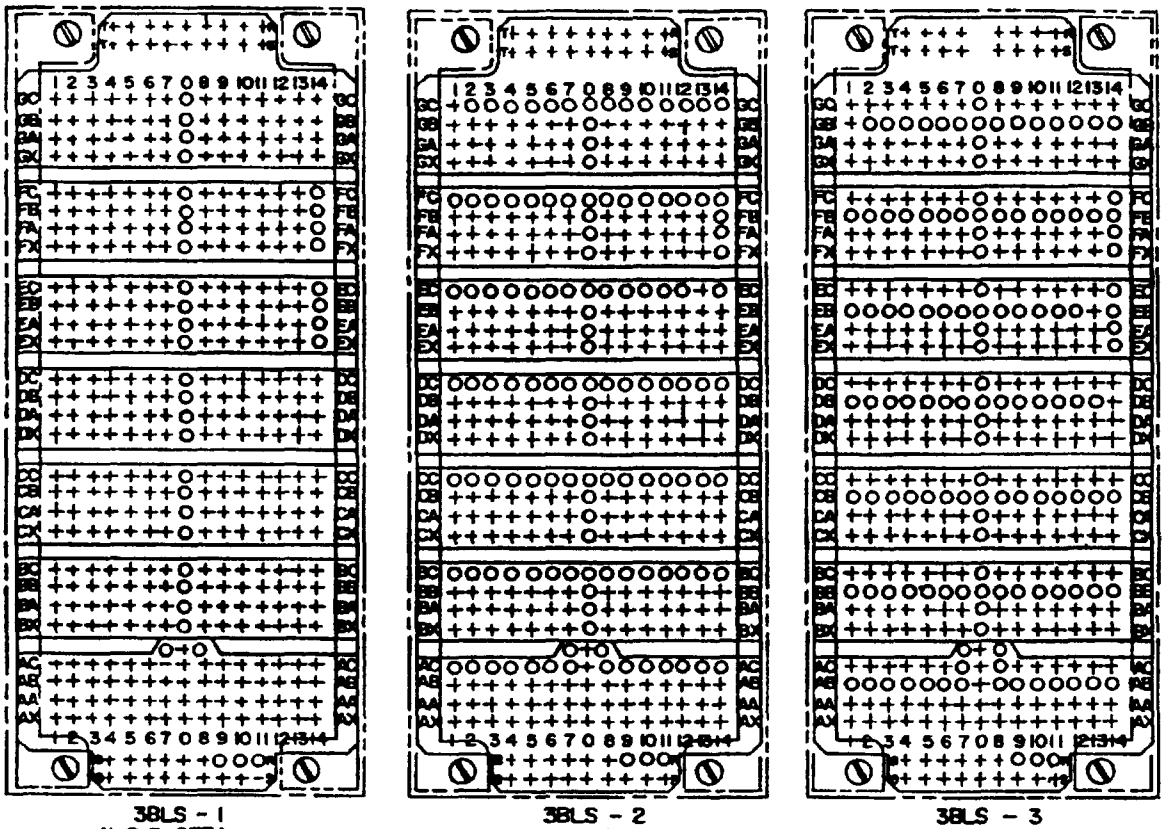
TABLE III. *Stop positions versus switching action.*

Stop position		Switch position accessible	Switching action
MBLS	RBLS		
1	1 and 5	A-B-C-D	1-2-3-OFF 1-2-3
1 and 2	1 and 4	A-B-C	1-2-OFF OFF-1-2[1]
1 and 3	1 and 3	A-B	1-2 ON-OFF OFF-ON[1]

[1]With stops located as specified, a switch assembly shall provide two or three active positions, any of which can be used as an OFF position by omitting external connections to the terminals in the desired OFF positions

Contact designation: Each stationary switching contact shall be identified by two letters and a number. The first letter, A, B, C, D, E, F, or G shall identify the pole group in which a contact is located and the second letter shall be assigned as shown on figure 2. The number shall be that of the column in which a contact is located (see figure 3). Each contact in the switch contact module connector plug shall be assigned the same identification as the contact in the switch contact module to which it connects. Contact D is common with contact X in all poles. Contacts shall be installed as shown on figure 3. For switch assemblies which include fuseholders (see table I), terminals shall be provided in the contact modules for connecting the fuseholder line, load and indicator circuits. Terminals shall be located as shown on figure 3. Terminal identification shall consist of the applicable row marker TR, TS, BR, BS, and the applicable column number, for example, TS 10.

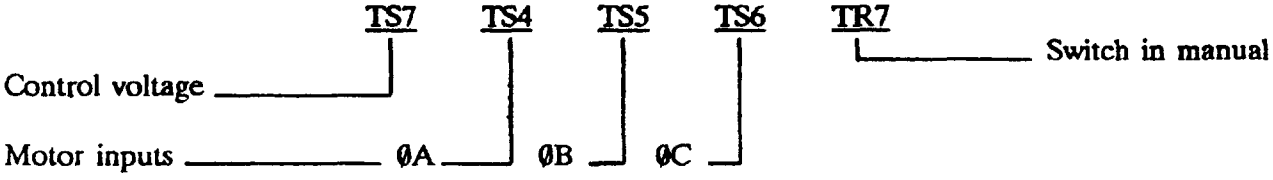
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NOTE. Bushings and contacts shall be installed in the stationary plate (803-1853221-21) as shown on figure 3. Crosses signify bushings and contacts which are installed. Circles signify omitted bushings and contacts. Bushings and contacts for fuseholder wiring shall be installed as required for the fuseholders installed (see tables I and V)

FIGURE 3. *Contact module marking, rear view.*

Remote operation: Connections to the switch module for remote operation shall be made to contacts as follows:



Remotely operated switch assembly shall move to the positions shown in table IV when the specified control voltage is applied.

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TABLE IV. *Control voltage for remote operation.*

Control voltage	Closed switch contacts	XFMR tap[1,2]
0	X-A	A
16.4	X-B	B
32.8	X-C	C
49.2	X-D	D

[1]See figure 3 of MIL-S-24187 for typical test circuits for remote-operated switch assemblies. Test loading is shown on figure 7 of this document.

[2]See figure 2 for typical pole positions corresponding to the transformer tap voltages A through D

Enclosure and wiring switch module:

Enclosure shall be furnished with the wiring module to connect the contact module to the ship's wiring connector receptacles at the rear of the enclosure (see figure 5). The general arrangement of the wiring shall be as shown on figure 4. Interconnecting wiring shall be as shown in tables V and VI.

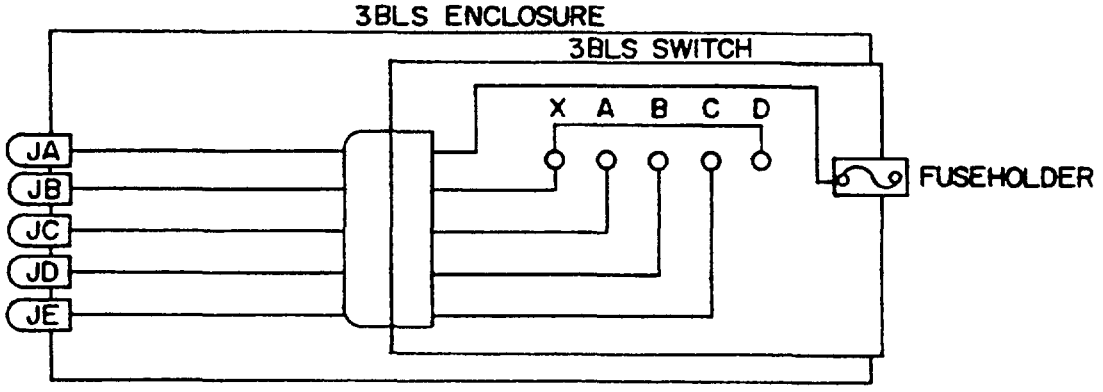


FIGURE 4. *Interconnecting wiring arrangement.*

Wiring module connections

For load switching circuits each 120 pin connector will be wired to a specific group of contacts of a switch contact module using the data shown in table V and figure 5. Wire shall be type B-20, multi-strand of MIL-W-16878. The color coding shall be as illustrated in the EXAMPLE below. For shield circuits 120 pin connector contacts [A] and [B] shall be connected with green wire to switch contacts F13 and G14. For other than switching circuits 120 pin connector contact [DH] shall be connected with gray wire to switch contact D13.

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TABLE V. *Contacts in switch contact module.*

Switch contact module contact														
	Column													
Pole[1]	1	2	3	4	5	6	7	8	9	10	11	12	13	14
G		L	M	N	Y	P	Z	AA	R	T	S	AM	AC	B
F	V	AE	W	AF	X	AH	AK	AJ	AB	AL	AX	AW	A	
E	AP	AR	AZ	BA	AS	AT	BC	BB	AU	AV	BD	BE		
D	BJ	BX	BT	BS	BL	BK	BM	BN	BU	BV	BP	BF	DH	
C	CE	CP	BZ	BY	CF	CH	CJ	CK	CA	CB	CL	CM	CX	CC
B	CR	CZ	DA	DB	CS	CT	DC	DN	CU	CV	DD	DE	DF	CW
A	DJ	DU	DV	DK	DL	DW	DM	DX	DY	DZ	DR	DP	EA	DS
	120 pin connector contact													

[1]Each pole contains contacts X, A, B, C and D (see figure 4)

Example: Pole E 5 that is connected to pin AS in accordance with table V is illustrated below for all connections:

Contact	Connector	Wire
X	JB	Black
A	JC	White
B	JD	Red
C	JE	Blue
D	None	None

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TABLE VI. *Wiring module connections, 38 pin connector (JA).*

Manual			Remote	
Fuse service	Switch contact	Connector JA contact	Switch contact	Service
BFI	TRO[1]	V	TRO[1]	BFI
BFI	TSO[2]	X	TSO[2]	BFI
1 LOAD	TR11	PP	TR11	–
1 LINE	TS11	RR	TS11	–
2 LOAD	TR10	JJ	TR10	–
2 LINE	TS10	KK	TS10	–
3 LOAD	TR9	DD	TR9	–
3 LINE	TS9	EE	TS9	–
4 LOAD	TR8	Z	TR8	–
4 LINE	TS8	AA	TS8	–
5 LOAD	TR7	R	TR7	Sw in manual
5 LINE	TS7	S	TS7	Control
6 LOAD	TR6	L	TR6	–
6 LINE	TS6	M	TS6	Phase C
7 LOAD	TR5	E	TR5	–
7 LINE	TS5	F	TS5	Phase B
8 LOAD	TR4	A	TR4	–
8 LINE	TS4	B	TS4	Phase A
9 LOAD	BR7	LL	EX-14	
9 LINE	BS7	SS	EA-14	SWITCH
10 LOAD	BR6	FF	EB-14	POLES
10 LINE	BS6	MM	EC-14	USED
11 LOAD	BR5	N	FX-14	FOR
11 LINE	BS5	J	FA-14	SYSTEM
12 LOAD	BR4	H	FB-14	CONTROL
12 LINE	BS4	C	FC-14	
BFI	BRO[1]	JUMPER TO TRO	BRO	–
BFI	BSO[2]	JUMPER TO TSO	BSO	–
	GX1	T	GX1	
	GA1	P	GA1	
	GB1	K	GB1	
	GC1	D	GC1	
SWITCH	EX13	U	EX13	SWITCH
POLES	EA13	W	EA13	POLES
USED	EB13	Y	EB13	USED
FOR	EC13	CC	EC13	FOR
SYSTEM	DX14	BB	DX14	SYSTEM
CONTROL	DA14	HH	DA14	CONTROL
	DB14	NN	DB14	
	DC14	TT	DC14	

[1]Return, blown fuse indication
[2]Supply, blown fuse indication

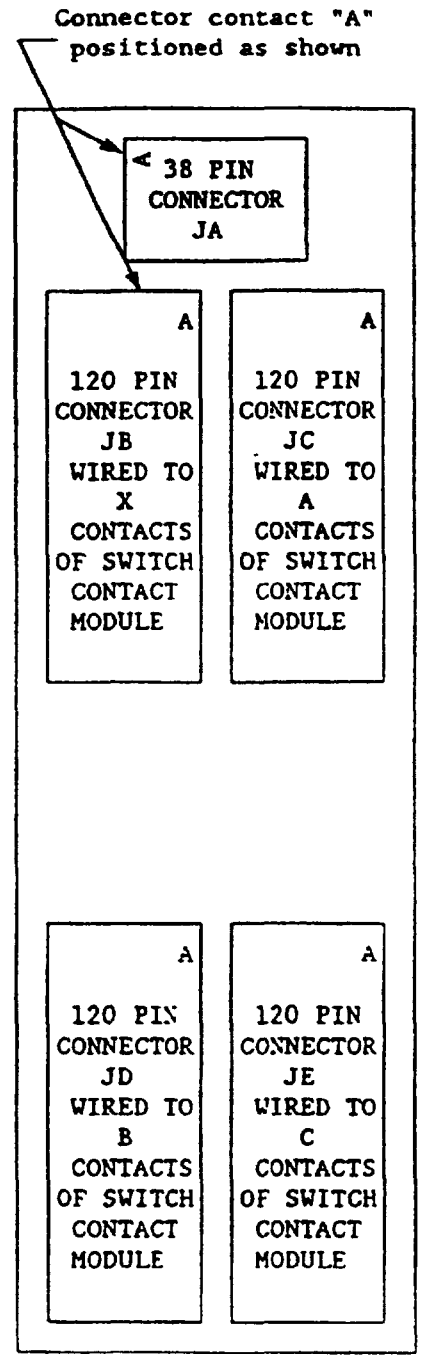


FIGURE 5. *Enclosed switch connector arrangements*

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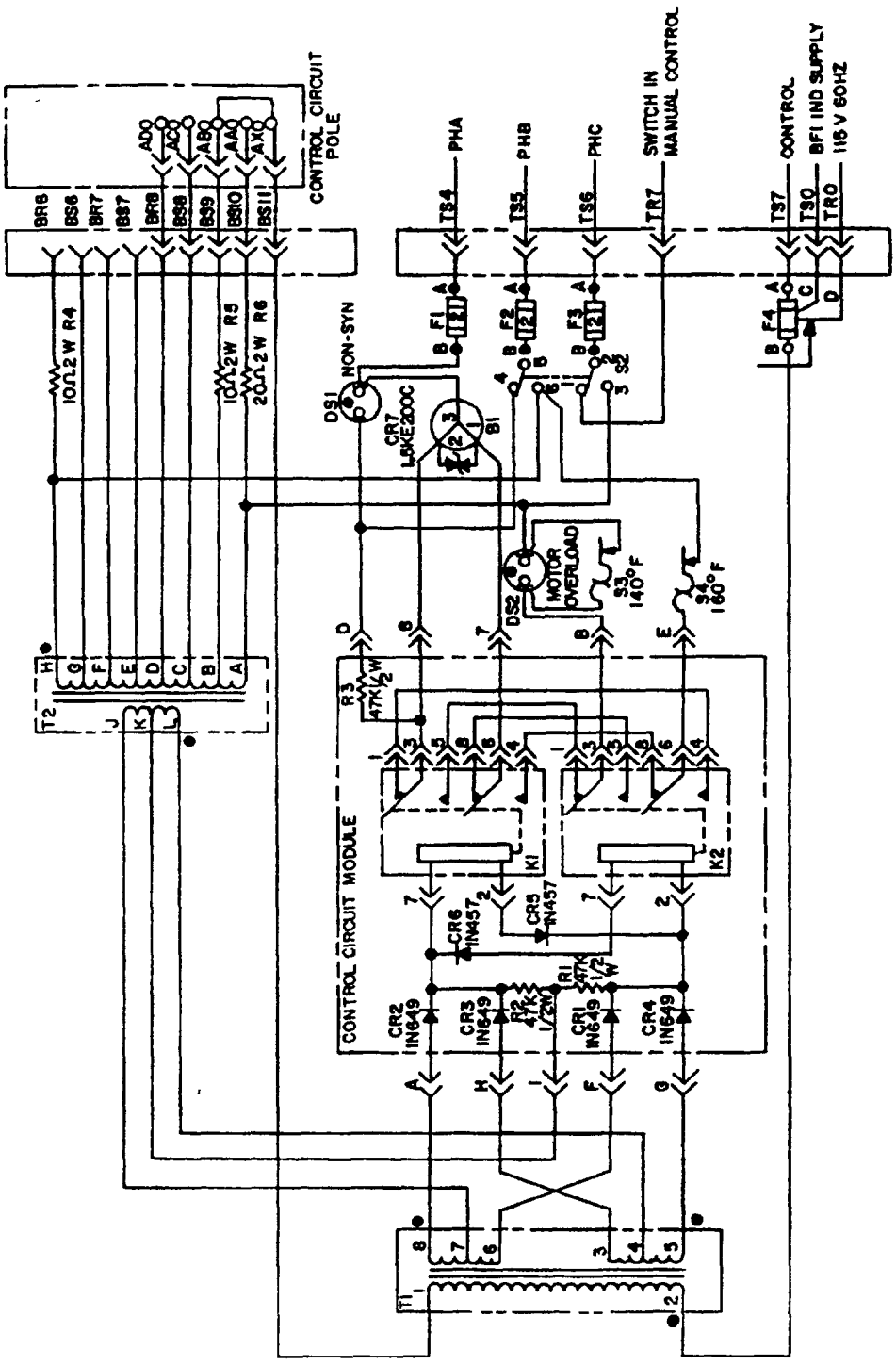


FIGURE 6. *Remote control circuit.*

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Repair parts. Repair parts shall be as specified in table VII.

TABLE VII. Repair parts.

Drawing 803-1853221 piece No.		Description	Quantities required for quantity of switches installed					
			1–3	4–12	13–30	31–60	61–120	121 or more
Manual	Remote							
1, 11, 13		Enclosed switch, without fuseholders	1	1	1	1	2	2
2, 7, 8		Enclosed switch, with fuseholders	1	1	1	1	2	2
	14, 15	Enclosed switch	1	1	1	1	2	2
7		Switch, type M3BLS6-1A	1	1	1	1	2	2
B		Switch, type M3BLS6-2A	1	1	1	1	2	2
2		Switch, type M3BLS6-3A	1	1	1	1	2	2
11		Switch, type M3BLS0-1B	1	1	1	1	2	2
13		Switch, type M3BLS0-2B	1	1	1	1	2	2
1		Switch, type M3BL60-3B	1	1	1	1	2	2
	14	Switch, type R3BLS0-1C	1	1	1	1	2	2
	15	Switch, type R3BLS0-2C	1	1	1	2	2	3
	16	Switch, type R3BLS0-3C	1	1	1	2	3	6
108	108	Stop screw	1	1	1	2	2	3
39	39	Detent spring	1	1	1	2	3	6
	87	Toggle switch	1	1	1	2	3	4
	85	Indicator light	1	1	1	2	3	5
	86	Indicator light	1	1	1	2	3	5
		Lamp (M15098/11-001)	Note 2					
	10	Control circuit module	1	1	1	2	2	3
	93	Relay	1	2	3	Note 3		
	160	Washer, insul.	2	2	2	4	6	8
	159	Spacer, insul	2	2	2	4	6	8
	81	Fuseholder	1	1	1	2	3	4
		Fuse carrier (1A0981)	1	1	2	3	4	5
	82	Fuse, 2.0 ampere	Note 4					
	83	Fuse, 1.0 ampere	Note 4					
A/R		Fuse, load circuit	Note 4					

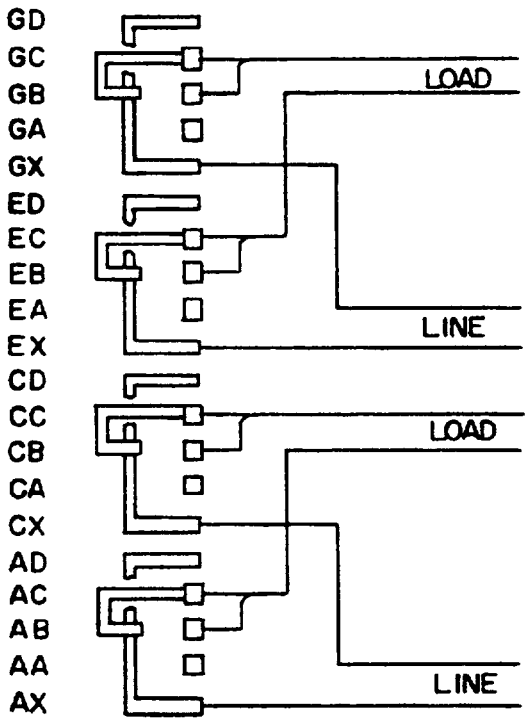
NOTES
1. Refer to Drawing 803-4680149 for piece numbers.
2. One hundred percent repair lamps shall be provided.
3. Supply one for each 10 or fraction of 10 installed.
4. One hundred fifty percent repair fuses shall be supplied. Fuses shall be supplied in multiples of five, in standard commercial packages, with a minimum quantity of 15.
5. Connector plugs, piece number 90 and 120, supplied only when specified.

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Qualification samples: Two type M3BLS6-1A or type R3BLS0-1C switch assemblies shall be submitted for qualification inspection (RBLS qualifies MBLS; MBLS does not qualify RBLS).

Tests:

Loading of switches: Loading of switches for endurance and temperature rise tests for qualification and quality conformance, group C testing shall be as shown on figure 7 and table VIII.



COLUMN	VOLTS	AMPERES	POWER FACTOR	Hz
1	115	4	1.0	60
4	115	3	0.3 INDUCTIVE	60
7	115	3	0.1 INDUCTIVE	400
11	28	4	RESISTIVE	dc
14	1.5	25 μA	RESISTIVE	dc

FIGURE 7 *Test loading*

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Temperature rise: Contacts to be monitored during temperature rise test are specified in table VIII.

TABLE VIII. *Temperature measurements.*

Column	Contacts
1	GX, GB
1	CX, CB

Revision letters are not used to denote changes due to the extensiveness of the changes.

Preparing activity:
Navy – SH
(Project 5930-N681-03)

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